

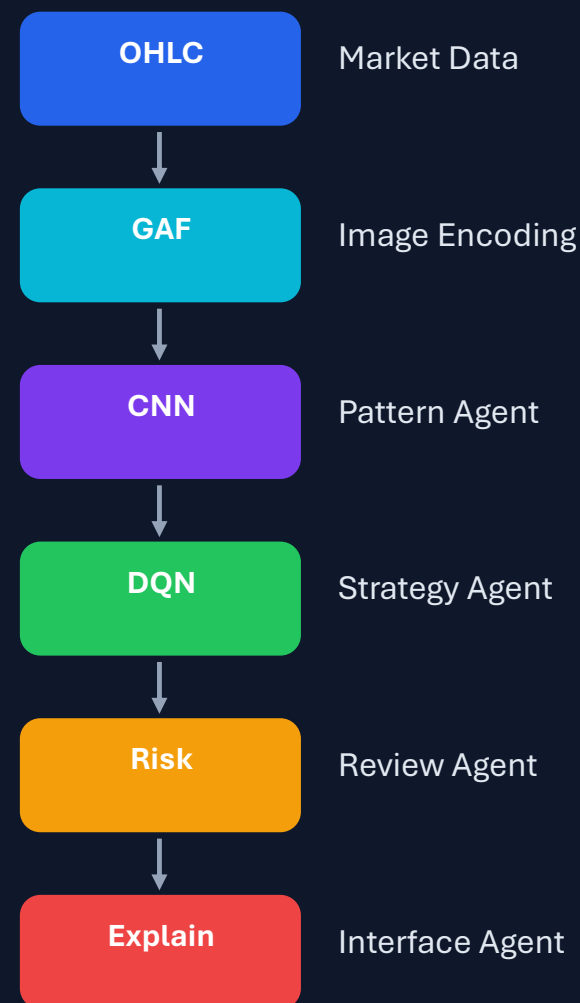
From Pattern Recognition to Agentic Decision Making

A 15-minute teaching demo using UW_AgenticAI
GAF-CNN + DQN + Risk Review + Explanation

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Core message: Agentic AI is workflow-level intelligence — models with roles, checks, and explanation.



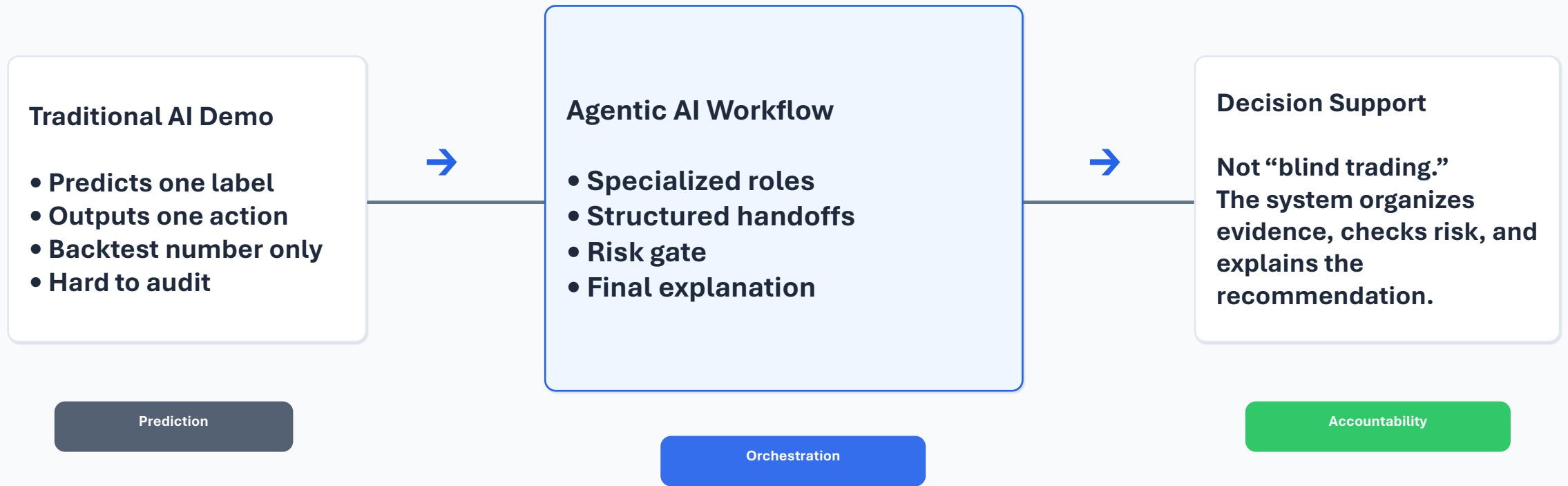


https://peculab.github.io/dqn_policy_explained_decisions.html

How does this system make and explain decisions?

Why this is more than “a trading model”

Frame the repository as an Agentic AI teaching case, not just CNN + DQN.



Show how each model becomes an agent-like responsibility in a decision chain.

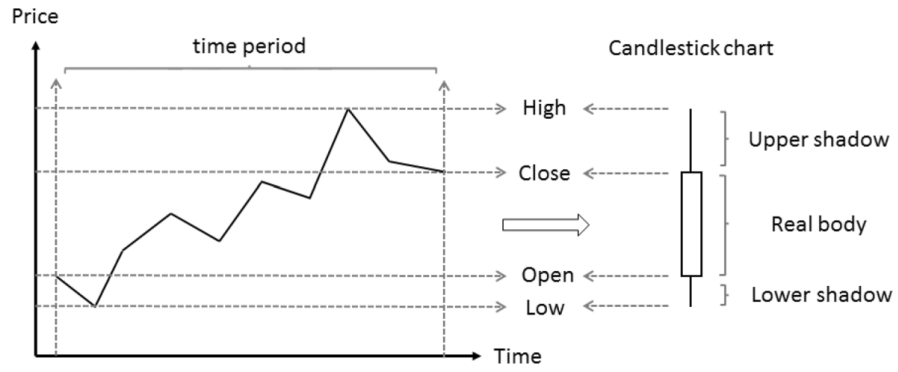
System pipeline: from OHLC to recommendation

The main story is a chain of evidence, not a single black-box prediction.



Teaching emphasis
you can inspect each boundary: input → representation
→ model output → risk gate → human-readable answer.

Agentic emphasis
The pipeline creates structured context for
strategy, risk, and explanation agents.



Our first step to making a GAF matrix is to normalize the given time series data X into values between $[0,1]$. The following equation shows the simple linear normalization method, where notation $\tilde{x}_i$ represents the normalized data.

$$\tilde{x}_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \quad (1)$$

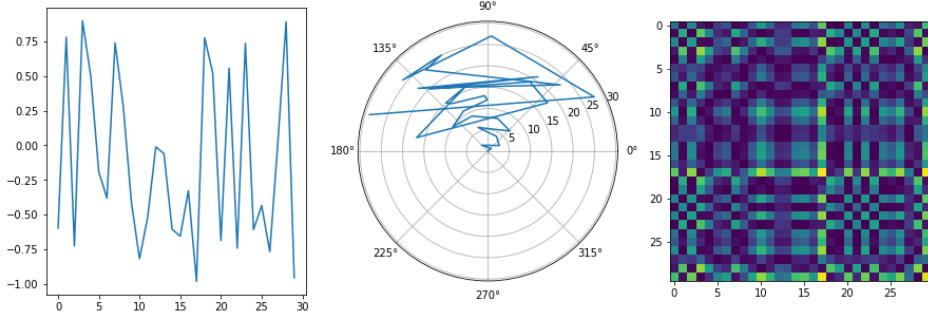
After normalization, our second step is to represent the normalized time series data in the polar coordinate system. The following two equations show how to get the angles and radius from the rescaled time series data.

$$\phi = \arccos(\tilde{x}_i), -1 \leq \tilde{x}_i \leq 1, \tilde{x}_i \in \tilde{X} \quad (2)$$

$$r = \frac{t_i}{N}, t_i \in \mathbb{N} \quad (3)$$

Finally, we sum the angles and use the cosine function to make the GASF by the following equation:

$$\text{GASF} = \cos(\phi_i + \phi_j) = \tilde{X}^T \cdot \tilde{X} - \sqrt{I - \tilde{X}^2}^T \cdot \sqrt{I - \tilde{X}^2} \quad (4)$$



Window ending at 2026-05-05 09:00:00+00:00 | derived bias: neutral

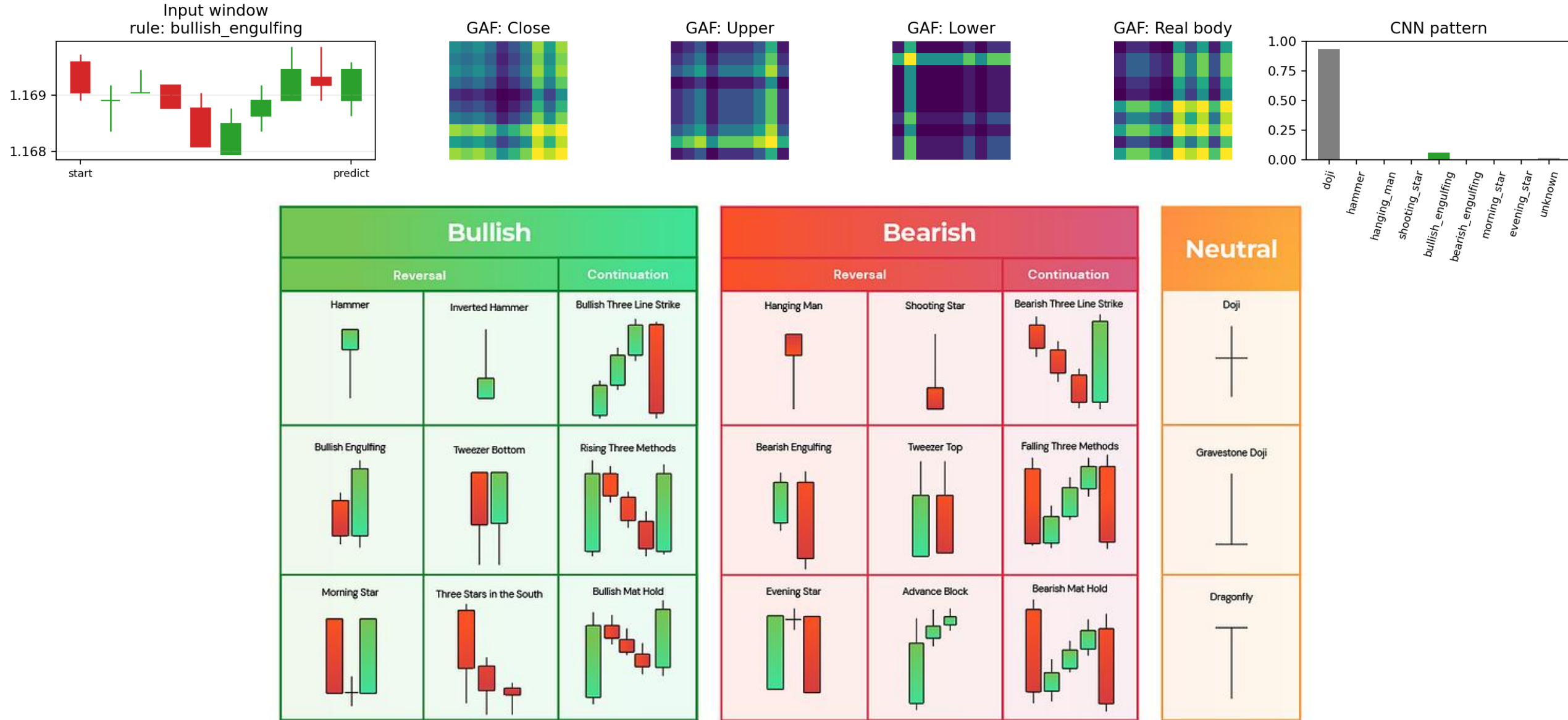
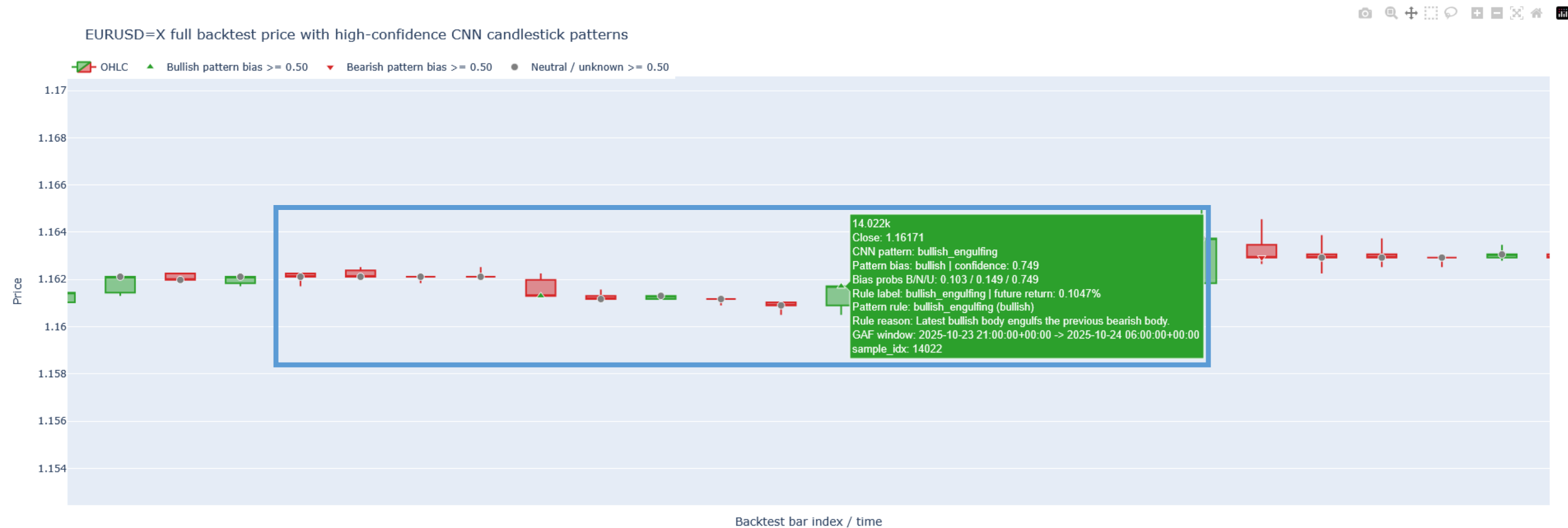


Figure Ref. <https://kongmunist.medium.com/candlestick-charts-are-bullshit-5906f3ae87b4>

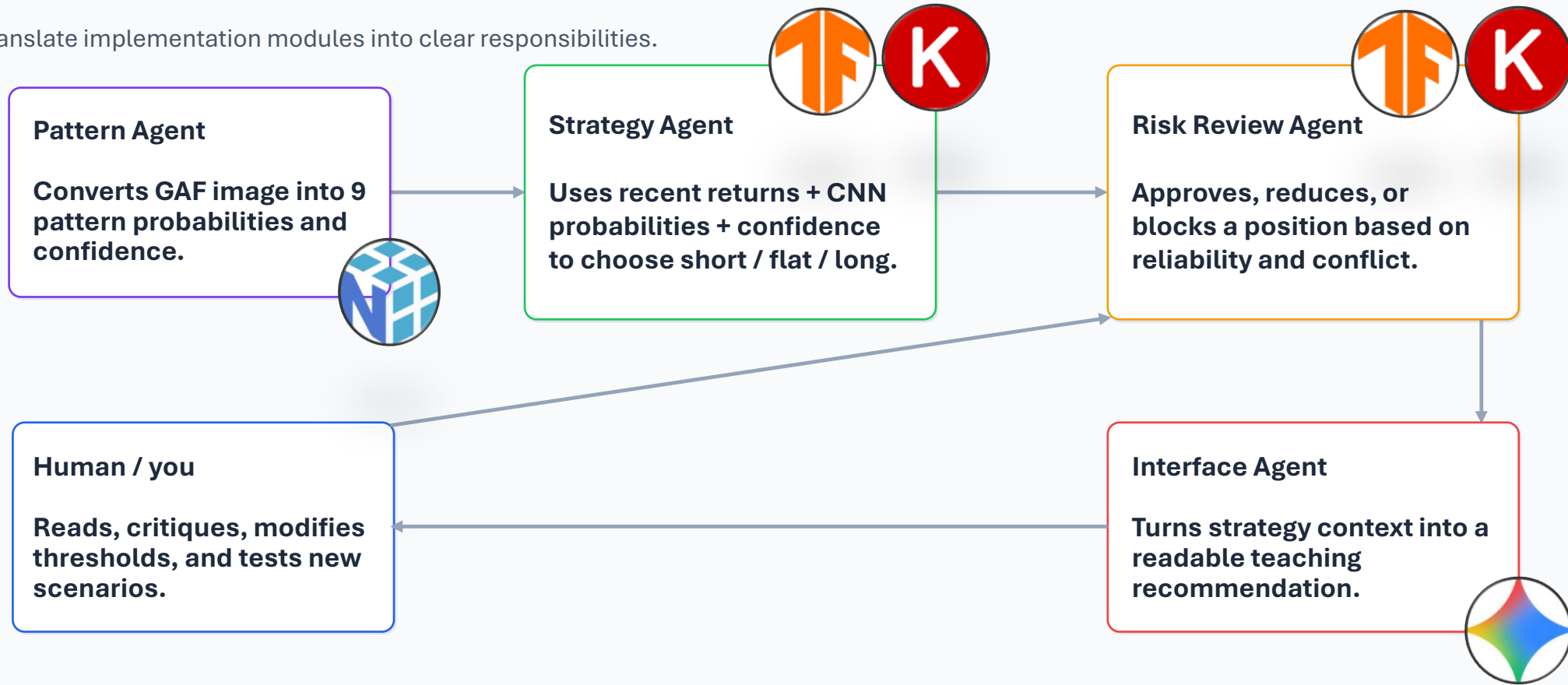


https://peculab.github.io/cnn_signal_price_chart.html

What does the Pattern Agent see?

Agent roles in UW_AgenticAI

Translate implementation modules into clear responsibilities.



Agentic AI is not only an LLM chatbot. It is a design pattern for roles, evidence, checks, and action boundaries.

Demo flow: what to show live

Keep the live demo small: input setting, agentic context, and final recommendation.

```
TICKER    = "EURUSD=X"  
PERIOD    = "730d"  
INTERVAL  = "1h"  
WINDOW    = 10
```



```
strategy =  
agentic_context["latest_dqn_strategy"]  
risk      =  
agentic_context["latest_risk_review"]  
final     =  
agentic_context["final_recommendation"]
```



```
print(strategy["action"])  
print(strategy["q_values"])  
print(risk["decision"])  
print(risk["suggested_position_s  
ize"])
```

Live output snapshot

Action: long / short / flat

Risk: approve / reduce / block

Position: +1.0, +0.5, 0, -0.5, -1.0

Explanation: a concise final recommendation

Teaching trick

Do not over-run the notebook. Show one successful result, then ask: “Where should we put the human control point?”

https://github.com/peculab/UW_AgenticAI/blob/main/Trading_AgenticAI_GAF_CNN_DQN_Demo.ipynb

You exercise: turn the demo into learning

Each exercise changes one system boundary and observes the downstream effect.

1

Change market

EUR/USD vs. GBP/USD vs.
AUD/USD

2

Change window

How does GAF image size
change the representation?

3

Change threshold

How often does the risk gate
reduce or block?

4

Tune DQN

Compare return, drawdown, and
trade count.

5

Modify rules

When do CNN pattern rules
conflict with DQN?

Pedagogical goal

You learn that Agentic AI is a controllable system: change the evidence, policy, review, or explanation, then measure what changes.

Takeaway

In the Agentic AI era, the question is no longer only:

“Can the model predict?”

The better question is:

**“Can the system organize evidence,
check risk, and explain its decision?”**

<https://scholar.google.com/citations?user=a2LHNL8AAAAJ>

UW_AgenticAI: a compact classroom demo for agentic decision-support workflows.

GitHub: [peculab/UW_AgenticAI](https://github.com/peculab/UW_AgenticAI)

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